

Newton's Laws

Building Models to Describe our World

Newton, Forces, and Free Body Diagrams



Forces

- There is no such thing as a force.
- These are a mathematical tool that allows us to describe the motion of objects.
- We represent them with a vector, as they have a magnitude and a direction.
- We can use a spring scale to define the magnitude and direction of a force.
- One issue is that we have some intuition about what a force feels like; it's better to apply rules for modelling forces, rather than trying to guess and apply intuition. Some concepts are counter-intuitive.
 - With practice, you will develop intuition about whether to include a specific force, not about what a force really is. The “physics intuition” is about getting comfortable applying the rules, not about understanding them. Nobody “understands” the rules.

Group exercise: Draw FBD for each object



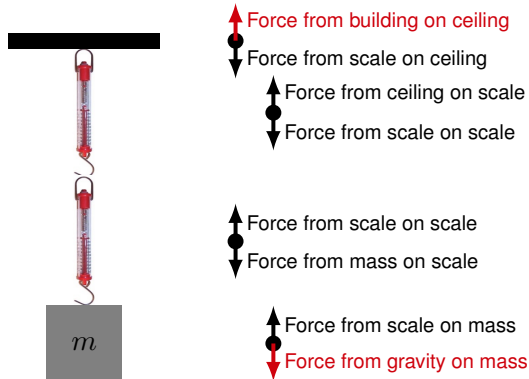
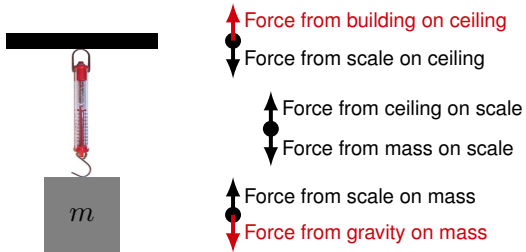
- ceiling
- scale
- mass



- ceiling
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- scale
- mass

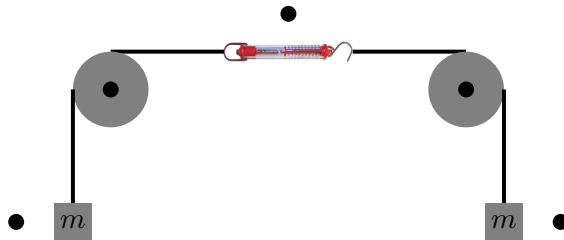
- Assume the scales are massless.
- Draw free body diagrams for the masses, scales and ceiling in both cases. (Draw little arrows around each point!).
- Remember, all objects must have an acceleration of zero!
- Be clear about what exerts a force on what!

Group exercise: Draw FBD for each object



- Two forces on each object, and they must have same magnitude/opposite direction.
- The Earth and building (which holds up the ceiling) exert forces but are not shown. We only show bodies on which forces are exerted.
- You need to become comfortable identifying forces that are exerted on objects!

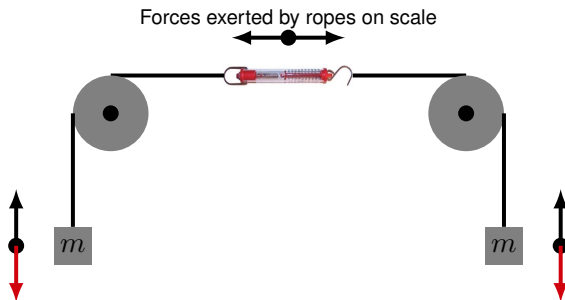
Group exercise: Draw the FBDs for masses and scale



Two masses, two pulleys and a spring scale.

Group exercise: Draw the FBDs for masses and scale

- All forces have the same magnitude, but the pulleys change the direction of tension forces in the ropes.
- The forces exerted on the scale are the same as in the previous problem, so it reads the same!
- The Earth exerts the two forces of gravity (in red).



Two masses, two pulleys and a spring scale

Newton's first law

An object will remain in its state of motion, be it at rest or moving with constant velocity, unless a net external force is exerted on the object.

If the above is true, then you're observing the object from an inertial frame of reference. Think of this as the way to define an inertial frame of reference.

Newton's Second Law

$$\sum \vec{F} = m\vec{a}$$

- This statement is a quantitative framework that describes almost all of the world around us. It leads to the concepts of conservation of energy, momentum, angular momentum, etc. It is an amazing equation!
- It connects dynamics (forces) to kinematics (acceleration).
- It is a vector equation $\rightarrow$ To be “useful” we need to write it out in components.

Newton's third law

If one object exerts a force on another object, the second object exerts a force on the first object that is equal in magnitude and opposite in direction.

This is not about intuition!!! **This is a rule to be followed!** (science)

If you include a force on object A that is exerted by object B, then you need to include an equal and opposite force exerted on object B from object A (if you care about modeling the motion of object B).